Hardik Santosh Gaonkar

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AI & ML undergraduate who builds AI systems end to end, from on-device LLM inference to RAG, ML and computer vision, and proves they work with measured benchmarks, held-out evaluation and automated tests. Seeking AI/ML engineering internships.

EDUCATION

BMS Institute of Technology and Management, Bengaluru — 2024 to 2028 (expected)
B.E. in Artificial Intelligence and Machine Learning — CGPA: **8.09** / 10

Class XII (PUC), **St. Joseph College of Commerce and Science**: **93.6%** · Class X, **Balmandir High School**: **97.6%**

PROJECTS

MachLume Studio | Hardware-aware AI profiler desktop app for Windows (CPU, GPU, NPU)

Rust · Tauri 2 · TypeScript · Preact · Windows PDH / DXCore / RAPL · Intel IGCL · OpenVINO · Ollama · llama.cpp

- Built a native Windows app (Tauri 2 + ~7K-line Rust core) that reads live hardware telemetry with no admin rights or drivers: per-core CPU and clocks, package, core and iGPU power (RAPL), page faults, thermal throttling, GPU engines (IGCL) and NPU load (DXCore).
- Detected NPUs (Intel AI Boost, AMD Ryzen AI, Qualcomm Hexagon) through DXCore and Plug and Play, and benchmarked them via OpenVINO, checked by Windows' NPU counters: **3.13 TOPS** FP16 on Intel AI Boost (Ollama used it 0 ms).
- Profiled local LLM runs in Ollama, splitting load, prefill and decode on the telemetry clock, and benchmarked llama.cpp's 4-bit x86 GEMM, GEMV, dot-product and f16 attention kernels, ported to Rust and checked against a scalar reference.
- Predicted every run with a roofline model from the model's tensor table, measured ceilings and llama.cpp's config; on 16 Llama 3 8B runs (42 to 1,652 prompt tokens), leave-one-out calibrated error was **2.4% prefill and 3.8% decode** (pure model 4.8% / 9.6%).
- Turned measurements into advice: showed Llama 3 8B decode is memory-bound (12.5 tok/s = 54.5 GB/s, the measured bandwidth) and 14 threads speed prefill 35% but not decode; **54 Rust tests** run on real counters, and the UI may call only 18 native commands.

CareFlow AI | Clinical AI platform with access-controlled RAG, ML and X-ray triage — github.com/hrdk6/CareFlow-AI

Python · FastAPI · PostgreSQL + pgvector · scikit-learn · XGBoost · SHAP · ONNX Runtime · Next.js · TypeScript · Docker

- Built a full-stack clinical AI platform where the LLM sees only data the user may access (RBAC + row-level SQL filters), and a 3-pass PII gateway (vault, regex, BERT NER) masks patient identities before cloud LLM calls.
- Implemented hybrid RAG (bge-small embeddings + BM25, RRF fusion, cross-encoder reranking, citation validation), reaching **Recall@1 0.955 and MRR 0.977** on a 48-question benchmark; reranking lifted context precision from 0.62 to 0.84.
- Trained explainable risk models on 99K patient encounters: a calibrated Random Forest for 30-day readmission (**PR-AUC 2× base rate**, SHAP) and XGBoost length of stay, with patient-grouped splits and leakage ablations.
- Applied transfer learning to chest X-ray triage: logistic heads on frozen ResNet-50 (ONNX) features, trained on 26K NIH ChestX-ray14 films; **8 findings reached ROC-AUC 0.71 to 0.80** on 7.5K patient-disjoint test films, with class activation maps.

TECHNICAL SKILLS

Languages	Python, TypeScript, Rust, Go, JavaScript, SQL
GenAI & LLMs	RAG, hybrid search (BM25 + dense), reranking, embeddings, LLM agents, LangGraph, LLM evaluation, guardrails, on-device inference (llama.cpp, Ollama, OpenVINO, NPUs), roofline performance modelling; Claude, OpenAI, Gemini
ML, NLP & CV	scikit-learn, XGBoost, SHAP, pandas, NumPy, Hugging Face, ONNX Runtime, NER, transfer learning
Backend & Web	FastAPI, Pydantic, PostgreSQL, pgvector, REST APIs, React, Next.js, Preact, Tauri 2 (desktop apps)
Infra & DevOps	Docker, Kubernetes, GitHub Actions (CI/CD), Git, Windows performance counters (PDH, DXCore, RAPL)

CERTIFICATIONS & ACHIEVEMENTS

- **Certified:** AI Engineer Core Track (LLM Engineering, RAG, QLoRA, Agents), Udemy, Oct 2026 (33.5 hours).
- Secured **3rd place** for an AI-based Task Scheduler at a competitive technical event; participated in multiple hackathons.